Florula Aramiensis

A Flora of the Arami Uplands

Volume I

Economic Trees

Edited by

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Arami Uplands Herbarium

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Note on the data
The Arami Uplands are a fictional region. The species treated in this volume are real West African trees, and their descriptions follow the published literature cited in each treatment. All specimen records, collectors' numbers, localities and dates are synthetic and were constructed for demonstration. They must not be cited as records of real collections.

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Preface

The Arami Uplands have no flora of their own. Collectors who worked in the region over the past century sent their specimens to herbaria in three different countries, and the names under which those specimens were filed have since been revised many times. A forester, an extension officer or a student who wants to identify a tree in the Uplands today has to work from regional floras that treat the area as a footnote.

This volume begins to close that gap. It treats the economic trees of the Uplands: the timber species, the fruit and seed trees that feed households and markets, and the trees whose bark, leaves or gums are used in medicine and craft. These are the species that people in the region most often need to name correctly, because a wrong name on a seedling order, a timber permit or a conservation list has real consequences.

The treatments are arranged by family, eight in this volume. Each family opens with a short description and a key to its genera and species. Each species treatment gives the accepted name and its principal synonyms, local names, a description, notes on phenology and habitat, the uses recorded in the literature, and the specimens on which the treatment rests.

Later volumes will cover the shrubs, climbers and herbs of the Uplands. Corrections and additional records are welcome and should be sent to the Arami Uplands Herbarium.

R. A.
Arami Uplands Herbarium
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How to use this Flora

Arrangement

Families are arranged alphabetically, and within each family the genera and species are also arranged alphabetically. The Key to Families (Chapter 3) leads to a family; the key at the start of each family chapter leads to a species.

Keys

The keys are dichotomous. Each couplet offers two contrasting leads, numbered **1a** and **1b**, **2a** and **2b**, and so on. Choose the lead that fits the specimen and follow the dotted line to its end, which gives either the number of the next couplet or a name. The short example below shows the layout only.

1a Leaves compound 2
1b Leaves simple ..... Key to Families
2a Leaves bipinnate; flowers in pendent red heads ..... *Parkia*
2b Leaves once pinnate; flowers in panicles ..... *Khaya*

Names

Scientific names are printed in italics and are followed by their author citation in roman type, for example *Milicia excelsa* (Welw.) C.C.Berg. Author abbreviations follow the standard forms listed in the Author Abbreviations appendix. Principal synonyms are given under each accepted name.

Specimen citations

Each specimen is cited by collector, collector's number and the herbarium in which it is kept, for example Alade 104 (AUH). Herbarium codes used in this volume are:

Code	Herbarium
AUH	Arami Uplands Herbarium
OKF	Okeri Forestry Research Institute Herbarium
SVN	Savanna Botanical Station Herbarium

Abbreviations

alt. altitude
c. about (*circa*)
fl. flowering
fr. fruiting
s.n. without number (*sine numero*)
syn. synonym

Chapter 1

Introduction

1.1 The Arami Uplands

The Arami Uplands occupy about 4,800 km^2 of dissected plateau between the Ireke River in the south and the Asiri River in the north. The land rises from about 350 m along the Ireke valley to 900 m on the Aramiwo Escarpment, the highest ground in the region. The main settlements are Okeri, the district headquarters and seat of the Okeri Forestry Research Institute, and Idofin, a market town on the northern plateau.

1.2 Climate

The Uplands have two wet seasons, from April to July and from September to October, separated by a short dry spell in August. The long dry season runs from November to March and is broken by the dry, dusty harmattan wind from the north. Mean annual rainfall falls from about 1,600 mm in the Ireke valley to about 1,200 mm on the northern plateau.

1.3 Vegetation

The region lies in the transition between the Guineo-Congolian rain forest and the Sudanian savanna woodlands described by White (1983). Three broad vegetation types are recognised in this Flora.

Lowland semi-deciduous forest covers the southern slopes below about 500 m. Its canopy trees include *Khaya grandifoliola*, *Entandrophragma cylindricum*, *Milicia excelsa* and *Terminalia superba*.

Forest–savanna mosaic occupies the middle slopes between about 500 and 700 m, where gallery forest along streams alternates with grassland that burns most years.

Guinea savanna woodland covers the northern plateau above about 700 m. Here the fire-tolerant *Vitellaria paradoxa*, *Parkia biglobosa* and *Pterocarpus erinaceus* are characteristic, and many of them are kept in farmland because of the value of their fruits and seeds.

1.4 History of botanical collecting

The first substantial collections from the Uplands were made by E. M. Harding between 1934 and 1938, during forest reserve surveys around Okeri. J. A. Olaniyan collected widely on the northern plateau between 1961 and 1967 for the Savanna Botanical Station. The most recent collections were made by the editor between 2019 and 2025 for this Flora. Specimens from all three periods are cited in the species treatments.

1.5 Scope and sources

This volume treats the economic trees of the Uplands. Descriptions draw on the regional floras of Hutchinson and Dalziel (1954), Aubréville (1959), Keay (1989) and Aké Assi (2001), and on the field guides of Arbonnier (2004) and Hawthorne and Jongkind (2006). Information on uses follows Burkill (1985), Louppe et al. (2008) and Orwa et al. (2009). Nomenclature follows the *Shenzhen Code* (Turland et al., 2018), and author abbreviations follow Brummitt and Powell (1992).

Chapter 2

Materials and Methods

2.1 Data files

The Flora rests on four tab-separated data files in the `data` folder of this project. They are the single source for every count, date and locality printed in this volume.

`data/species.tsv`
: The checklist: one row for each of the 18 species, with its family, author citation, status (native or cultivated), the vegetation types in which it occurs, and its local and trade names.

`data/localities.tsv`
: The 12 collecting localities, with grid square, altitude and vegetation type. It matches the Gazetteer in Appendix B row for row.

`data/specimens.tsv`
: The 90 specimens on which the treatments rest, with collector, collector's number, date, locality, altitude, herbarium and phenological state.

`data/phenology.tsv`
: For each species and month, whether the species is recorded in flower (fl), in fruit (fr), both, or neither.

2.2 Validation of the specimen records

Every specimen record was checked against six rules before it was accepted.

1. The altitude equals the altitude of the locality in `localities.tsv`.

2. According to `species.tsv`, the species occurs in the vegetation type of the locality.

3. The date falls within the collector's working period: Harding 1934–1938, Olaniyan 1961–1967, Alade 2019–2025.

4. The herbarium is the one that holds that collector's material: Harding in OKF, Olaniyan in SVN, Alade in AUH.

5. A specimen in flower or fruit was collected in a month in which `phenology.tsv` records that state; a sterile specimen was collected in a month with neither.

6. Each collector's numbers increase with date and are never repeated.

2.3 Summary of the specimen data

Table 2.1 gives the number of species and specimens in each family, and Table 2.2 gives the number of specimens from each collector.

Table 2.1: Species and specimens by family.

Family	Species	Specimens
Anacardiaceae	2	10
Combretaceae	2	10
Fabaceae	3	15
Irvingiaceae	1	5
Malvaceae	3	15
Meliaceae	3	15
Moraceae	2	10
Sapotaceae	2	10
Total	18	90

Table 2.2: Specimens by collector and phenological state.

Collector	Herbarium	Flowering	Fruiting	Sterile	Total
Harding	OKF	8	19	5	32
Olaniyan	SVN	11	16	5	32
Alade	AUH	4	12	10	26
Total		23	47	20	90

Chapter 3

Key to Families

This key covers only the eight families treated in this volume. Where a family keys out twice, the genus reached at that lead is given in brackets.

1a Leaves compound 2

1b Leaves simple 5

2a Leaves digitately compound, the leaflets all arising from the tip of the petiole ..... Malvaceae (*Adansonia*, *Ceiba*)

2b Leaves pinnate or bipinnate 3

3a Fruit a woody capsule splitting into 4 or 5 valves; seeds winged ..... Meliaceae

3b Fruit not a valved capsule 4

4a Leaves once pinnate with a terminal leaflet; crushed leaves with a resinous smell; fruit a fleshy drupe ..... Anacardiaceae (*Spondias*)

4b Leaves bipinnate, or pinnate with a swollen pulvinus at the base of the petiole; fruit a pod or a flat winged disc ..... Fabaceae

5a Slash exuding white latex 6

5b Slash without white latex 7

6a Stipules present, leaving a scar round the twig; flowers small, crowded in spikes or heads Moraceae

6b Stipules absent; leaves crowded at the ends of the twigs; fruit a berry with large glossy seeds Sapotaceae

7a Terminal bud enclosed in a long conical stipule that leaves a ring scar; fruit a large drupe Irvingiaceae

7b Terminal bud not enclosed in a conical stipule 8

8a Branches in tiers, with the leaves crowded at their ends; fruit dry and winged ..... Combretaceae

8b Branches not in tiers; fruit not winged 9

9a Fruit a kidney-shaped nut borne on a swollen, fleshy stalk ..... Anacardiaceae (*Anacardium*)

9b Fruit of 4 or 5 spreading follicles with large seeds; petiole swollen at both ends Malvaceae (*Cola*)

Appendix A

Author Abbreviations

Author abbreviations used in the names in this volume follow Brummitt and Powell (1992). Every abbreviation that appears in an author citation in this volume must be listed here.

Abbreviation	Author	Dates
A.Chev.	Auguste Jean Baptiste Chevalier	1873–1956
A.Juss.	Adrien-Henri de Jussieu	1797–1853
Aubry-Lecomte	Charles Eugène Aubry-Lecomte	1821–1898
Baill.	Henri Ernest Baillon	1827–1895
C.C.Berg	Cornelis Christiaan Berg	1934–2012
C.DC.	Anne Casimir Pyramus de Candolle	1836–1918
C.F.Gaertn.	Carl Friedrich von Gaertner	1772–1850
Desr.	Louis Auguste Joseph Desrousseaux	1753–1838
Diels	Friedrich Ludwig Emil Diels	1874–1945
Endl.	Stephan Friedrich Ladislaus Endlicher	1804–1849
Engl.	Heinrich Gustav Adolf Engler	1844–1930
G.Don	George Don	1798–1856
Gaertn.	Joseph Gaertner	1732–1791
Jacq.	Nikolaus Joseph von Jacquin	1727–1817
L.	Carl Linnaeus	1707–1778
Lesch.	Jean Baptiste Louis Théodore Leschenault de La Tour	1773–1826
O'Rorke	O'Rorke	not traced
Pers.	Christiaan Hendrik Persoon	1761–1836
Poir.	Jean Louis Marie Poiret	1755–1834
R.Br.	Robert Brown	1773–1858
Schott	Heinrich Wilhelm Schott	1794–1865
Sm.	James Edward Smith	1759–1828
Sprague	Thomas Archibald Sprague	1877–1958
Vent.	Étienne Pierre Ventenat	1757–1808
Welw.	Friedrich Martin Josef Welwitsch	1806–1872

Appendix B

Gazetteer of Localities

This gazetteer lists every locality cited in the specimen records of this volume. For reference, the Flora area is divided into grid squares: rows A to D run from north to south, and columns 1 to 5 run from west to east. Altitudes are in metres. Vegetation types follow the Introduction: LSF, lowland semi-deciduous forest (below 500 m); FSM, forest–savanna mosaic (500–700 m); GSW, Guinea savanna woodland (above 700 m).

Locality	Grid square	Altitude (m)	Vegetation
Agbede	C2	540	FSM
Aramiwo Escarpment, summit	B1	900	GSW
Aramiwo Foot	B2	640	FSM
Asiri Crossing	A2	740	GSW
Ewe Stream gallery forest	B3	580	FSM
Idofin	A3	760	GSW
Igbo-Oba Forest Reserve	C3	470	LSF
Ireke Ferry	D4	360	LSF
Oke-Asa farmland	B4	690	FSM
Okeri	C4	420	LSF
Okuta Hill	A4	820	GSW
Olowo Stream	D3	440	LSF

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